



Diffusion Models (Part 2)

<https://hku-data8018.github.io/>

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- Slides reference
 - MIT EECS Fall 2024, 6.S978 Deep Generative Models, <https://mit-6s978.github.io/>
 - <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>
 - <https://beckham.nz/2022/07/11/d3pms.html>
 - CMU 11-785 Introduction to Deep Learning, <https://deeplearning.cs.cmu.edu/F24/>

Diffusion Model

$$\mathcal{L} = \mathbb{E}_{x_0, t, \epsilon} \left[w_t \|\epsilon - \epsilon_\theta(x_t, t)\|^2 \right]$$

over p_{data}
 over $[1, T]$
 over $\mathcal{N}(0, \mathbf{I})$

Algorithm 1 Training

- 1: **repeat**
- 2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
- 3: $t \sim \text{Uniform}(\{1, \dots, T\})$
- 4: $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 5: Take gradient descent step on $\nabla_\theta \|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2$
- 6: **until** converged

Algorithm 2 Sampling

- 1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 2: **for** $t = T, \dots, 1$ **do**
- 3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
- 4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
- 5: **end for**
- 6: **return** $\mathbf{x}_0$

estimated μ

sampling from
estimated distribution

Diffusion Model (Score Matching)

$$\underbrace{D_F(q(\tilde{x}) \parallel p_\theta(\tilde{x}))}_{\text{Fisher divergence of noised data}} = \underbrace{\mathbb{E}_{q(x, \tilde{x})}}_{\text{joint distribution}} \left[\frac{1}{2} \parallel \underbrace{\nabla_{\tilde{x}} \log q(\tilde{x} \mid x)}_{\text{score of conditional}} - \underbrace{\nabla_{\tilde{x}} \log p_\theta(\tilde{x})}_{\text{parameterized score}} \parallel^2 \right] + \text{constant}$$

Algorithm 1 Training

```

1: repeat
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$ 
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$ 
4:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
5:   Take gradient descent step on
        $\nabla_\theta \parallel \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \parallel^2$ 
6: until converged
  
```

score function

Algorithm 2 Sampling

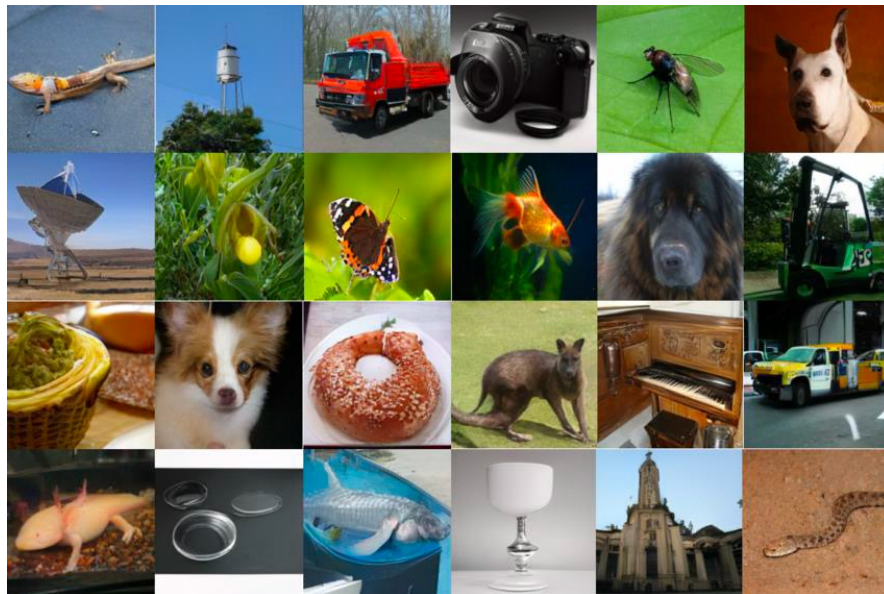
```

1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$ 
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ 
5: end for
6: return  $\mathbf{x}_0$ 
  
```

score function

Langevin Dynamics

Conditional Diffusion Generation



$$p(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$$



$$p(\mathbf{x}_{0:T} \mid y) = p(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1} \mid \mathbf{x}_t, y)$$

Conditional Diffusion Generation

- Using the perspective of **score matching**

$$\nabla_x \log p(x) = -\nabla_x E(x)$$

$$\begin{aligned}\nabla \log p(\mathbf{x}_t \mid y) &= \nabla \log \left(\frac{p(\mathbf{x}_t)p(y \mid \mathbf{x}_t)}{p(y)} \right) \\ &= \nabla \log p(\mathbf{x}_t) + \nabla \log p(y \mid \mathbf{x}_t) - \nabla \log p(y) \\ &= \underbrace{\nabla \log p(\mathbf{x}_t)}_{\text{unconditional score}} + \underbrace{\nabla \log p(y \mid \mathbf{x}_t)}_{\text{adversarial gradient}}\end{aligned}$$

Conditional Diffusion Generation

$$\nabla \log p(\mathbf{x}_t \mid y) = \nabla \log p(\mathbf{x}_t) + \gamma \nabla \log p(y \mid \mathbf{x}_t)$$

- **Classifier Guidance:** Training a classifier for

$$\nabla \log p(y \mid \mathbf{x}_t)$$

Conditional Diffusion Generation

$$\nabla \log p(\mathbf{x}_t \mid y) = \nabla \log p(\mathbf{x}_t) + \gamma \nabla \log p(y \mid \mathbf{x}_t)$$

- Classifier Guidance: Training a classifier for

$$\nabla \log p(y \mid \mathbf{x}_t)$$

- Limitation: depending on the performance of a separate classifier

Conditional Diffusion Generation

$$\nabla \log p(\mathbf{x}_t | y) = \nabla \log p(\mathbf{x}_t) + \gamma \nabla \log p(y | \mathbf{x}_t)$$

- Classifier Guidance: Training a classifier for
- **Classifier-free Guidance:**

$$\nabla \log p(y | \mathbf{x}_t) = \nabla \log p(\mathbf{x}_t | y) - \nabla \log p(\mathbf{x}_t)$$

$$\begin{aligned} \nabla \log p(\mathbf{x}_t | y) &= \nabla \log p(\mathbf{x}_t) + \gamma (\nabla \log p(\mathbf{x}_t | y) - \nabla \log p(\mathbf{x}_t)) \\ &= \nabla \log p(\mathbf{x}_t) + \gamma \nabla \log p(\mathbf{x}_t | y) - \gamma \nabla \log p(\mathbf{x}_t) \\ &= \underbrace{\gamma \nabla \log p(\mathbf{x}_t | y)}_{\text{conditional score}} + \underbrace{(1 - \gamma) \nabla \log p(\mathbf{x}_t)}_{\text{unconditional score}} \end{aligned}$$

Conditional Diffusion Generation

$$\nabla \log p(\mathbf{x}_t | y) = \nabla \log p(\mathbf{x}_t) + \gamma \nabla \log p(y | \mathbf{x}_t)$$

- Classifier Guidance: Training a classifier for
- **Classifier-free Guidance:**
 - Training a conditional diffusion model with **an additional unconditional class**

$$\mathbb{E}_{\mathcal{E}(x), y, \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_{\theta}(z_t, t, \tau_{\theta}(y))\|_2^2 \right]$$

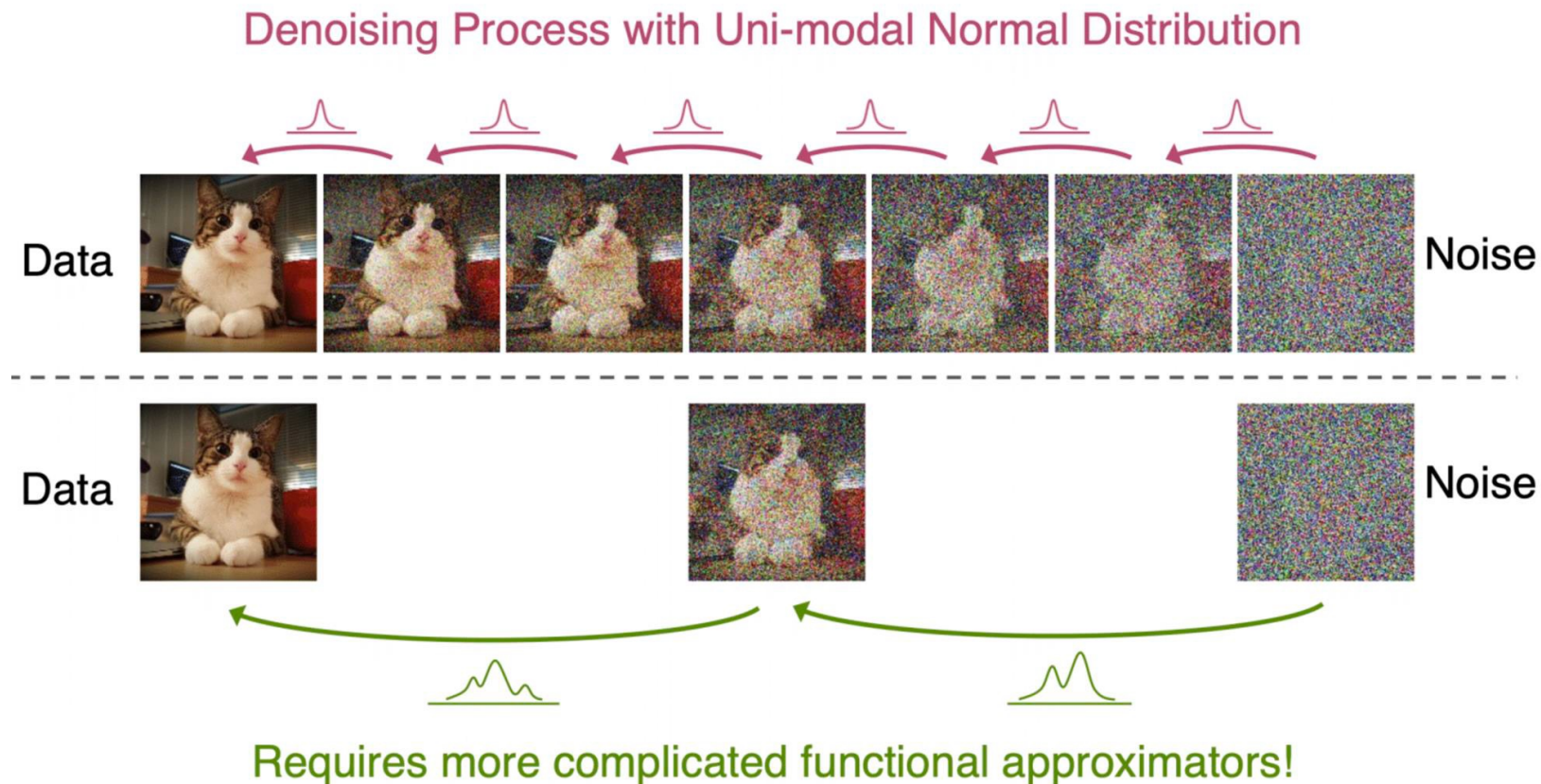
$$\underbrace{\gamma \nabla \log p(\mathbf{x}_t | y)}_{\text{conditional score}} + \underbrace{(1 - \gamma) \nabla \log p(\mathbf{x}_t)}_{\text{unconditional score}}$$

Speed-up Diffusion Model

- Generating a sample requires **T denoising steps** ($T = 1000$ in vanilla diffusion model)
 - 50k images of size 32 x 32
 - 20 hours for DDPM
 - < 1 min for GAN

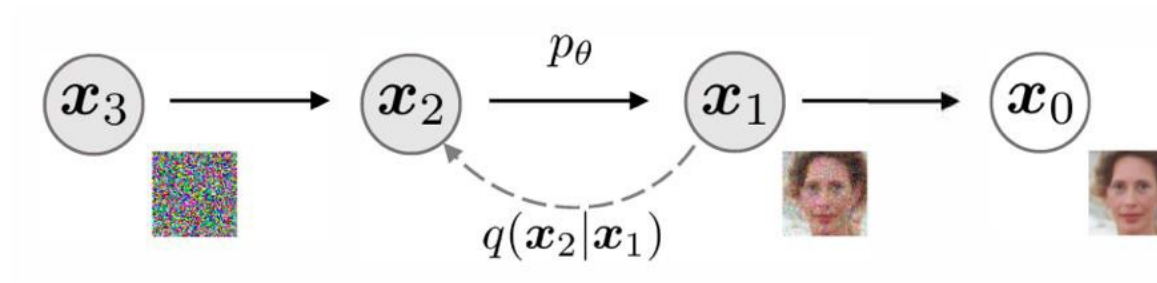
Speed-up Diffusion Model

- Skip some steps



Speed-up Diffusion Model

- Skip some steps: **Denoising Diffusion Implicit Model (DDIM)**



$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

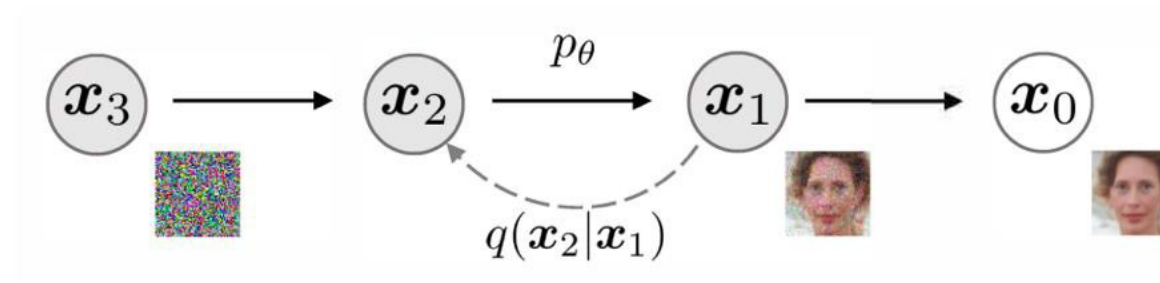
$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

$$L_{\text{simple}} = \mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), t \sim \mathcal{U}(1, T)} [\| \epsilon - \underbrace{\epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)}_{\mathbf{x}_t} \|^2]$$

Denoising Diffusion Probabilistic Model (DDPM)

Speed-up Diffusion Model

- Skip some steps: **Denoising Diffusion Implicit Model (DDIM)**



*only depends on
previous step*

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

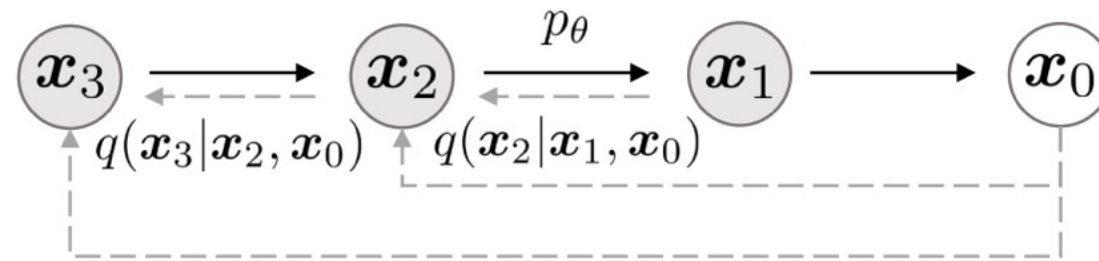
$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

$$L_{\text{simple}} = \mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), t \sim \mathcal{U}(1, T)} [\| \epsilon - \underbrace{\epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)}_{\mathbf{x}_t} \|^2]$$

Denoising Diffusion Probabilistic Model (DDPM)

Speed-up Diffusion Model

- Skip some steps: **Denoising Diffusion Implicit Model (DDIM)**



$$q_{\sigma}(\mathbf{x}_{1:T} \mid \mathbf{x}_0) := q_{\sigma}(\mathbf{x}_T \mid \mathbf{x}_0) \prod_{t=2}^T q_{\sigma}(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)$$

$$q_{\sigma}(\mathbf{x}_T \mid \mathbf{x}_0) = \mathcal{N}(\sqrt{\alpha_T} \mathbf{x}_0, (1 - \alpha_T) \mathbf{I})$$

$$q_{\sigma}(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\alpha_{t-1}} \mathbf{x}_0 + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\alpha_t} \mathbf{x}_0}{\sqrt{1 - \alpha_t}}, \sigma_t^2 \mathbf{I}\right)$$

Denoising Diffusion Implicit Model (DDIM)

Speed-up Diffusion Model

- Skip some steps: **Denoising Diffusion Implicit Model (DDIM)**

$$q_{\sigma}(\mathbf{x}_{1:T} \mid \mathbf{x}_0) := q_{\sigma}(\mathbf{x}_T \mid \mathbf{x}_0) \prod_{t=2}^T q_{\sigma}(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)$$

$$q_{\sigma}(\mathbf{x}_T \mid \mathbf{x}_0) = \mathcal{N}(\sqrt{\alpha_T} \mathbf{x}_0, (1 - \alpha_T) \mathbf{I})$$

$$q_{\sigma}(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\alpha_{t-1}} \mathbf{x}_0 + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\alpha_t} \mathbf{x}_0}{\sqrt{1 - \alpha_t}}, \sigma_t^2 \mathbf{I}\right)$$

- A **non-Markovian** forward process

$$q_{\sigma}(\mathbf{x}_t \mid \mathbf{x}_{t-1}, \mathbf{x}_0) = \frac{q_{\sigma}(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) q_{\sigma}(\mathbf{x}_t \mid \mathbf{x}_0)}{q_{\sigma}(\mathbf{x}_{t-1} \mid \mathbf{x}_0)}$$

Denoising Diffusion Implicit Model (DDIM)

Speed-up Diffusion Model

- Skip some steps: **Denoising Diffusion Implicit Model (DDIM)**

$$q_{\sigma}(\mathbf{x}_{1:T} \mid \mathbf{x}_0) := q_{\sigma}(\mathbf{x}_T \mid \mathbf{x}_0) \prod_{t=2}^T q_{\sigma}(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)$$

$$q_{\sigma}(\mathbf{x}_T \mid \mathbf{x}_0) = \mathcal{N}(\sqrt{\alpha_T} \mathbf{x}_0, (1 - \alpha_T) \mathbf{I})$$

$$q_{\sigma}(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\alpha_{t-1}} \mathbf{x}_0 + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\alpha_t} \mathbf{x}_0}{\sqrt{1 - \alpha_t}}, \sigma_t^2 \mathbf{I}\right)$$

- Backward process

$$p_{\theta}^{(t)}(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \begin{cases} \mathcal{N}(f_{\theta}^{(1)}(\mathbf{x}_1), \sigma_1^2 \mathbf{I}) & \text{if } t = 1 \\ q_{\sigma}(\mathbf{x}_{t-1} \mid \mathbf{x}_t, f_{\theta}^{(t)}(\mathbf{x}_t)) & \text{otherwise,} \end{cases}$$

$$f_{\theta}^{(t)}(\mathbf{x}_t) := (\mathbf{x}_t - \sqrt{1 - \alpha_t} \cdot \epsilon_{\theta}^{(t)}(\mathbf{x}_t)) / \sqrt{\alpha_t}$$

Denoising Diffusion Implicit Model (DDIM)

Speed-up Diffusion Model

- Skip some steps: **Denoising Diffusion Implicit Model (DDIM)**

Algorithm DDPM Sampling

$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for all t from T to 1 **do**
 $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
 $\mu \leftarrow \frac{1}{\sqrt{\alpha_t}}(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}}\epsilon_{\theta}(\mathbf{x}_t, t))$
 $\mathbf{x}_{t-1} \leftarrow \mu + \boxed{\sigma_t \epsilon}$ **Stochastic**
end for
return $\mathbf{x}_0$

Algorithm DDIM Sampling

$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for all t from T to 1 **do**
 $\bar{\epsilon} \leftarrow \epsilon_{\theta}(\boxed{\mathbf{x}_t}, t)$
 $\bar{\mathbf{x}}_0 \leftarrow \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t}\bar{\epsilon}}{\sqrt{\bar{\alpha}_t}}$ **Estimate $\mathbf{x}_0$**
 $\boxed{\mathbf{x}_{t-1}} \leftarrow \sqrt{\bar{\alpha}_{t-1}}\bar{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t-1}}\bar{\epsilon}$
end for
return $\mathbf{x}_0$

DDPM vs DDIM

Speed-up Diffusion Model

- Skip some steps: **Denoising Diffusion Implicit Model (DDIM)**

DDIM

Algorithm Original DDIM Sampling

```

 $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
for all  $t$  from  $T$  to 1 do
     $\bar{\epsilon} \leftarrow \epsilon_{\theta}(\mathbf{x}_t, t)$ 
     $\bar{\mathbf{x}}_0 \leftarrow \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \bar{\epsilon}}{\sqrt{\bar{\alpha}_t}}$ 
     $\mathbf{x}_{t-1} \leftarrow \sqrt{\bar{\alpha}_{t-1}} \bar{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t-1}} \bar{\epsilon}$ 
end for
return  $\mathbf{x}_0$ 
    
```

Increasing
Sub-sequence

$$[1, \dots, T] \implies [\tau_0 = 0, \dots, \tau_S = T]$$

E.g., $\tau = [0, 10, 20, 30, \dots, 1000]$

Algorithm Fewer-Steps DDIM Sampling

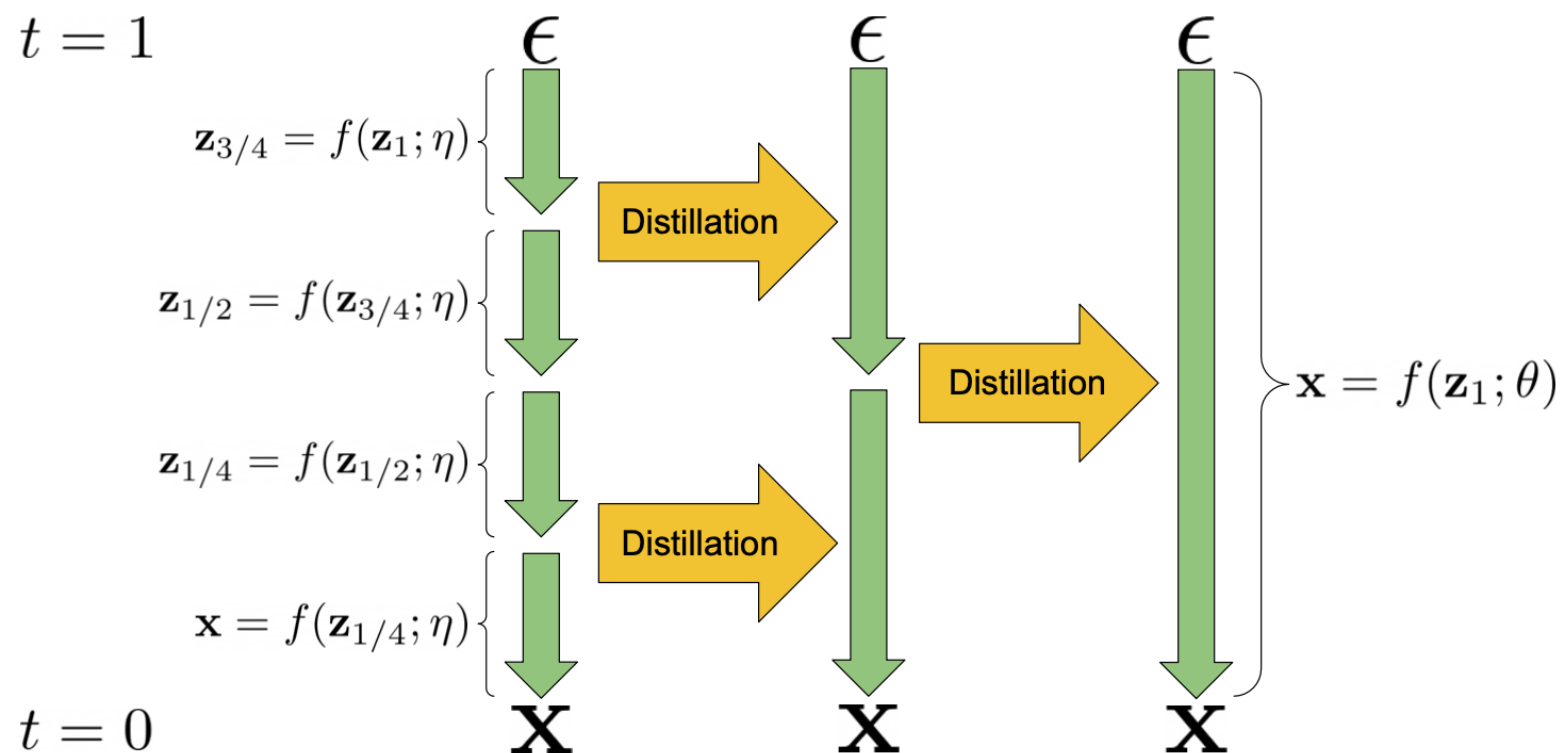
```

 $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
for all  $s$  from  $S$  to 1 do
     $t \leftarrow \tau_s$ 
     $t' \leftarrow \tau_{s-1}$ 
     $\bar{\epsilon} \leftarrow \epsilon_{\theta}(\mathbf{x}_t, t)$ 
     $\bar{\mathbf{x}}_0 \leftarrow \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \bar{\epsilon}}{\sqrt{\bar{\alpha}_t}}$ 
     $\mathbf{x}_{t'} \leftarrow \sqrt{\bar{\alpha}_{t'}} \bar{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t'}} \bar{\epsilon}$ 
end for
return  $\mathbf{x}_0$ 
    
```

DDIM with fewer steps

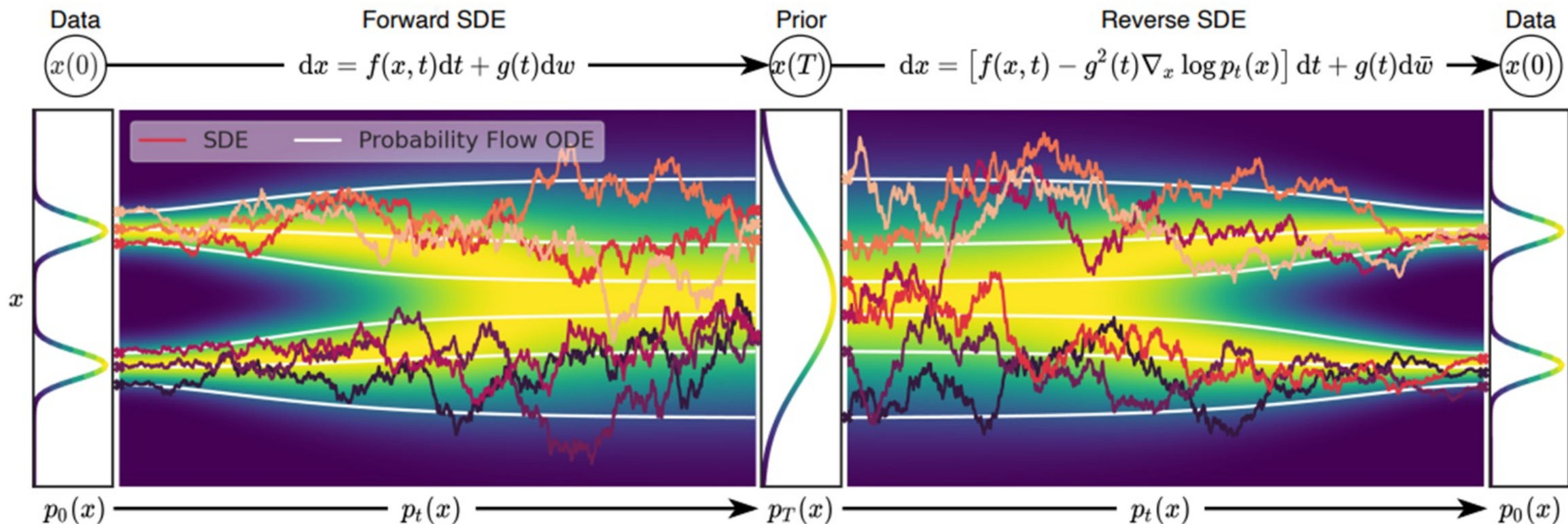
Speed-up Diffusion Model

- Skip some steps: Denoising Diffusion Implicit Model (DDIM)
- **Progressive distillation**



Speed-up Diffusion Model

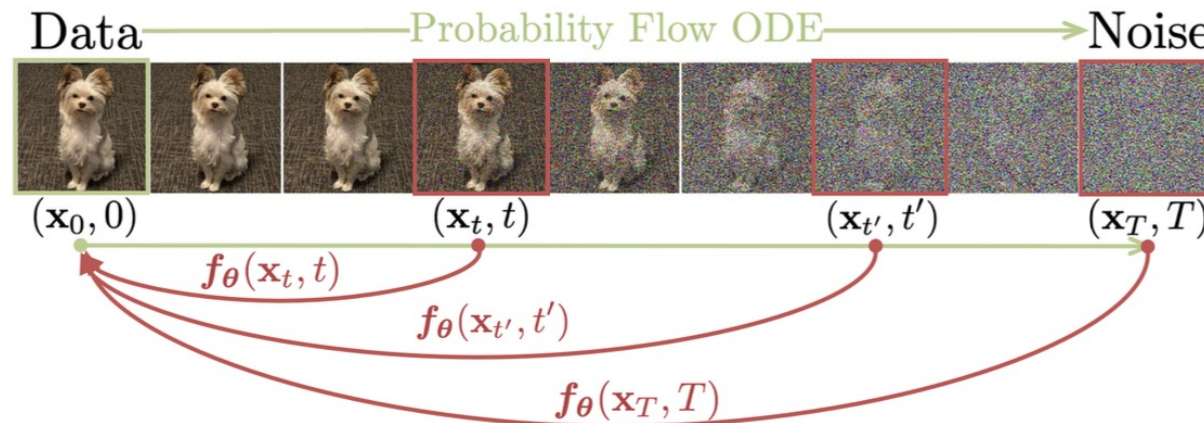
- Skip some steps: Denoising Diffusion Implicit Model (DDIM)
- Progressive distillation
- **Consistency model**



Speed-up Diffusion Model

- Skip some steps: Denoising Diffusion Implicit Model (DDIM)
- Progressive distillation
- **Consistency model**
 - Map from a point on any trajectory to the trajectory's origin

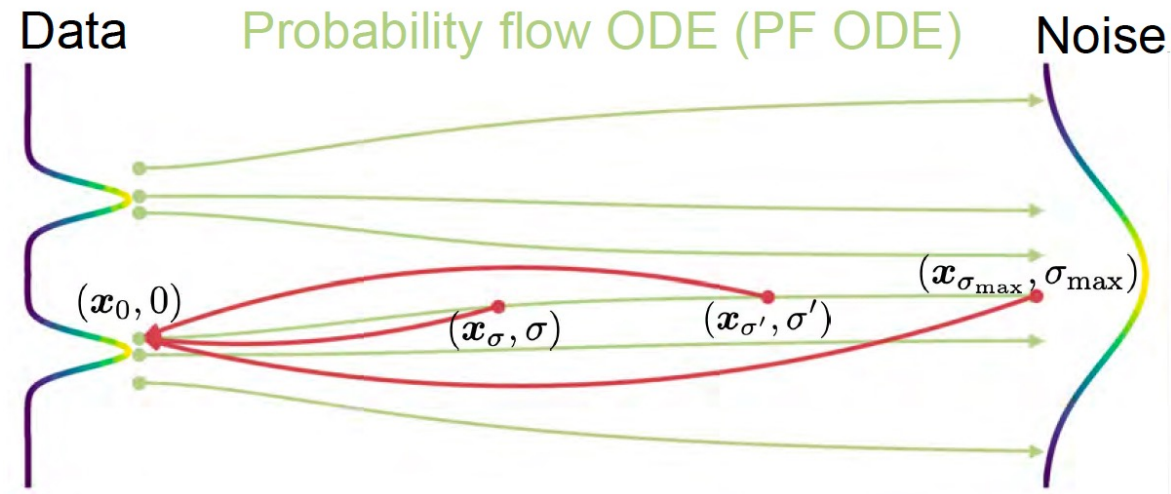
$$f_{\theta}(x_{\sigma}, \sigma) = x_0$$



Speed-up Diffusion Model

- Skip some steps: Denoising Diffusion Implicit Model (DDIM)
- Progressive distillation
- **Consistency model**
 - Map from a point on any trajectory to the trajectory's origin

$$f_{\theta}(x_{\sigma}, \sigma) = x_0$$



Speed-up Diffusion Model

- Skip some steps: Denoising Diffusion Implicit Model (DDIM)
- Progressive distillation
- **Consistency model**
 - Map from a point on any trajectory to the trajectory's origin

$$f_{\theta}(x_{\sigma}, \sigma) = x_0$$

Boundary condition

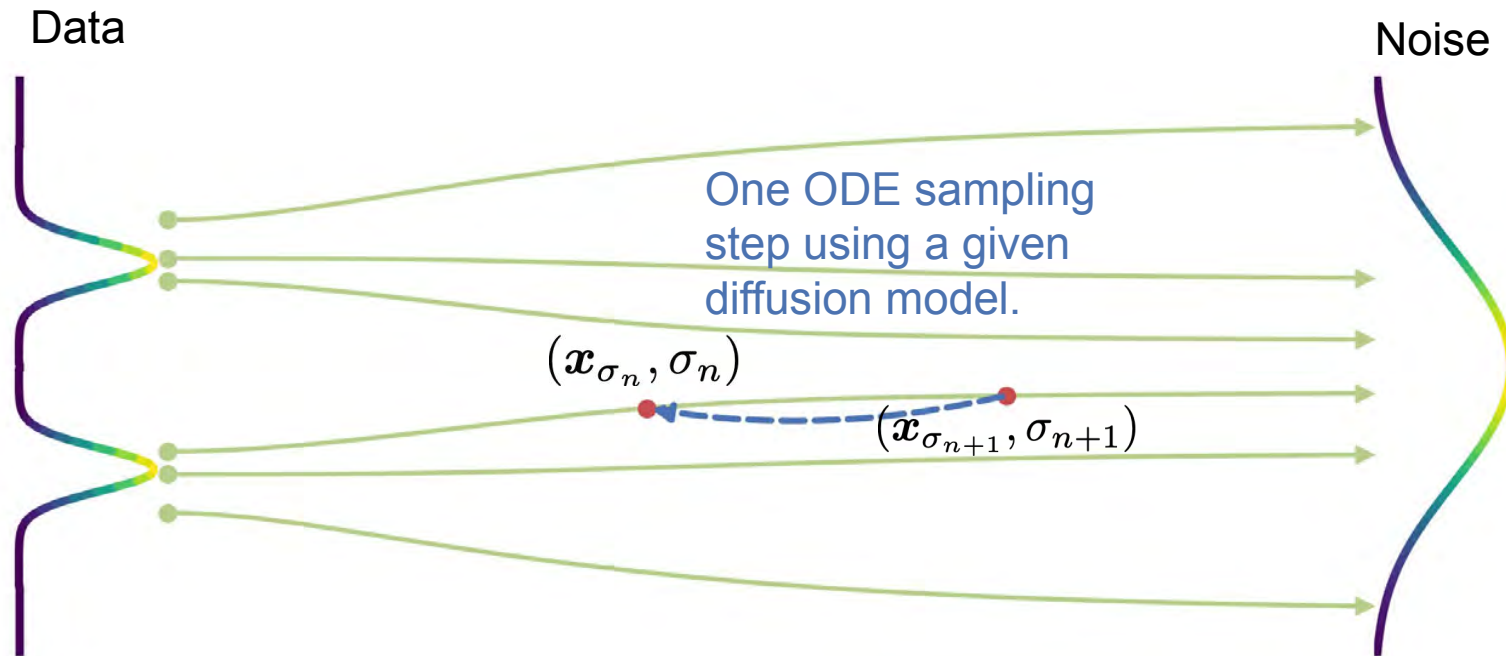
$$f_{\theta}(\mathbf{x}_0, 0) = \mathbf{x}_0$$

Self-consistency

$$\forall \sigma, \sigma' \in [0, \sigma_{\max}] : f_{\theta}(x_{\sigma}, \sigma) = f_{\theta}(x_{\sigma'}, \sigma')$$

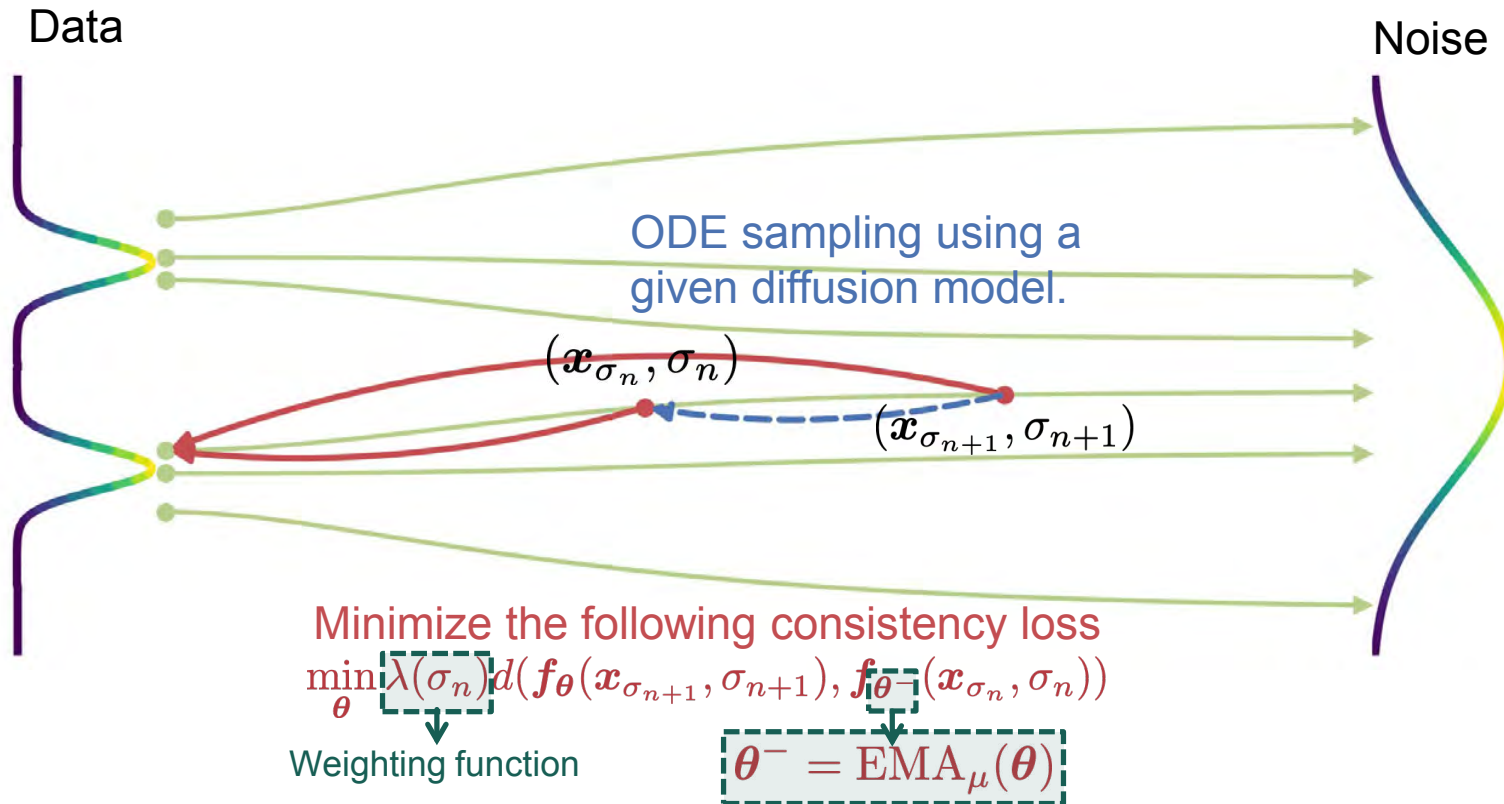
Speed-up Diffusion Model

- Skip some steps: Denoising Diffusion Implicit Model (DDIM)
- Progressive distillation
- **Consistency model**
 - Training: **Consistency Distillation**



Speed-up Diffusion Model

- Skip some steps: Denoising Diffusion Implicit Model (DDIM)
- Progressive distillation
- **Consistency model**
 - Training: **Consistency Distillation**



Speed-up Diffusion Model

- Skip some steps: Denoising Diffusion Implicit Model (DDIM)
- Progressive distillation
- **Consistency model**
 - Training: Consistency Distillation
 - Training: **Consistency Training**
 - Sample random noise level, data point, and Gaussian noise

$$\min_{\theta} \mathbb{E}[\lambda(\sigma_n) d(\mathbf{f}_{\theta}(\mathbf{x} + \sigma_{n+1}\mathbf{z}, \sigma_{n+1}), \mathbf{f}_{\theta}(\mathbf{x} + \sigma_n\mathbf{z}, \sigma_n))]$$

Diffusion Model for Discrete Data

- Language, protein, ...
- x_0 : clean data, x_T : noise. Finite state space of size M

Diffusion Model for Discrete Data

- Language, protein, ...
- $\mathbf{x}_0$: clean data, $\mathbf{x}_T$: noise. Finite state space of size M
- Forward process:
 - Iterative transition follows a transition matrix
- Marginal distribution at step t :

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \text{Cat}(\mathbf{x}_t; \mathbf{p} = \mathbf{x}_{t-1} \mathbf{Q}_t)$$
$$[\mathbf{Q}_t]_{ij} = q(x_t = j | x_{t-1} = i).$$

$$q(\mathbf{x}_t | \mathbf{x}_0) = \text{Cat}(\mathbf{x}_t; \mathbf{p} = \mathbf{x}_0 \bar{\mathbf{Q}}_t), \quad \text{with} \quad \bar{\mathbf{Q}}_t = \mathbf{Q}_1 \mathbf{Q}_2 \dots \mathbf{Q}_t$$

Diffusion Model for Discrete Data

- Language, protein, ...
- x_0 : clean data, x_T : noise. Finite state space of size M
- Forward process:
 - Iterative transition follows a transition matrix
$$q(x_t|x_{t-1}) = \text{Cat}(x_t; \mathbf{p} = x_{t-1} \mathbf{Q}_t)$$
$$[\mathbf{Q}_t]_{ij} = q(x_t = j | x_{t-1} = i).$$
 - Marginal distribution at step t :

$$q(x_t|x_0) = \text{Cat}(x_t; \mathbf{p} = x_0 \bar{\mathbf{Q}}_t), \quad \text{with} \quad \bar{\mathbf{Q}}_t = \mathbf{Q}_1 \mathbf{Q}_2 \dots \mathbf{Q}_t$$

- **Assumption:**
 - Must converge to a **known stationary distribution π** as t approaches infinity
 - Reverse process starts from **π**

Diffusion Model for Discrete Data

- Language, protein, ...
- $\mathbf{x}_0$: clean data, $\mathbf{x}_T$: noise. Finite state space of size M
- Reverse process:
 - Two key distributions: $q(\mathbf{x}_0|\mathbf{x}_t)$ and $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{\sum_{\mathbf{x}_0} q(\mathbf{x}_{t-1}, \mathbf{x}_t, \mathbf{x}_0)}{q(\mathbf{x}_t)} \quad (7)$$

$$= \frac{\sum_{\mathbf{x}_0} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)q(\mathbf{x}_0|\mathbf{x}_t)q(\mathbf{x}_t)}{q(\mathbf{x}_t)} \quad (7b)$$

$$= \sum_{\mathbf{x}_0} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)q(\mathbf{x}_0|\mathbf{x}_t) \quad (7c)$$

$$= \mathbb{E}_{q(\mathbf{x}_0|\mathbf{x}_t)} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \quad (7d)$$

Diffusion Model for Discrete Data

- Language, protein, ...
- $\mathbf{x}_0$: clean data, $\mathbf{x}_T$: noise. Finite state space of size M
- Reverse process:
 - Two key distributions: $q(\mathbf{x}_0|\mathbf{x}_t)$ and $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$
 - $\mathbf{x}_t \in \{0, 1\}^k$ (for any t) is a **discrete label** represented as a one-hot vector
 - $\tilde{\mathbf{x}}_t \in [0, 1]^k$ is the **predicted variable** and is not one-hot
 - $Q_t \in [0, 1]^{k \times k}$ is a **stochastic matrix** whose rows sum to 1

$$\begin{aligned} q(X_{t-1} | X_t = \mathbf{x}_t, X_0 = \mathbf{x}_0) &= \frac{q(X_t = \mathbf{x}_t | X_{t-1}, X_0 = \mathbf{x}_0) q(X_{t-1} | X_0 = \mathbf{x}_0)}{q(X_t = \mathbf{x}_t | X_0 = \mathbf{x}_0)} \\ &= \frac{q(X_t = \mathbf{x}_t | X_{t-1}) q(X_{t-1} | X_0 = \mathbf{x}_0)}{q(X_t = \mathbf{x}_t | X_0 = \mathbf{x}_0)} \end{aligned}$$

Diffusion Model for Discrete Data

- Language, protein, ...
- $\mathbf{x}_0$: clean data, $\mathbf{x}_T$: noise. Finite state space of size M
- Reverse process:
 - Two key distributions: $q(\mathbf{x}_0|\mathbf{x}_t)$ and $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$
 - $\mathbf{x}_t \in \{0, 1\}^k$ (for any t) is a **discrete label** represented as a one-hot vector
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$$q(X_{t-1} | X_t = \mathbf{x}_t, X_0 = \mathbf{x}_0) = \frac{q(X_t = \mathbf{x}_t | X_{t-1}, X_0 = \mathbf{x}_0) q(X_{t-1} | X_0 = \mathbf{x}_0)}{q(X_t = \mathbf{x}_t | X_0 = \mathbf{x}_0)}$$

$$= \frac{q(X_t = \mathbf{x}_t | X_{t-1}) q(X_{t-1} | X_0 = \mathbf{x}_0)}{q(X_t = \mathbf{x}_t | X_0 = \mathbf{x}_0)}$$

$$\underbrace{q(X_t = \mathbf{x}_t | X_{t-1} = \mathbf{x}_{t-1})}_{1 \times 1} = \underbrace{[\mathbf{x}_{t-1} \quad Q_t]}_{\substack{1 \times k \quad k \times k}} \underbrace{\mathbf{x}_t^T}_{k \times 1}$$

Diffusion Model for Discrete Data

- Language, protein, ...
- $\mathbf{x}_0$: clean data, $\mathbf{x}_T$: noise. Finite state space of size M
- Reverse process:
 - Two key distributions: $q(\mathbf{x}_0|\mathbf{x}_t)$ and $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$
 - $\mathbf{x}_t \in \{0, 1\}^k$ (for any t) is a **discrete label** represented as a one-hot vector
 - $\tilde{\mathbf{x}}_t \in [0, 1]^k$ is the **predicted variable** and is not one-hot
 - $Q_t \in [0, 1]^{k \times k}$ is a **stochastic matrix** whose rows sum to 1

$$q(X_{t-1} | X_t = \mathbf{x}_t, X_0 = \mathbf{x}_0) = \frac{q(X_t = \mathbf{x}_t | X_{t-1}, X_0 = \mathbf{x}_0) q(X_{t-1} | X_0 = \mathbf{x}_0)}{q(X_t = \mathbf{x}_t | X_0 = \mathbf{x}_0)}$$

$$= \frac{q(X_t = \mathbf{x}_t | X_{t-1}) q(X_{t-1} | X_0 = \mathbf{x}_0)}{q(X_t = \mathbf{x}_t | X_0 = \mathbf{x}_0)}$$

$$\underbrace{q(X_t = \mathbf{x}_t | X_{t-1})}_{k \times 1} = \underbrace{\begin{bmatrix} \mathbf{I}_k & Q_t \end{bmatrix}}_{\substack{k \times k & k \times k}} \underbrace{\mathbf{x}_t^T}_{k \times 1} = Q_t \mathbf{x}_t^T$$

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$$\begin{aligned}
 q(X_{t-1}|X_t = \mathbf{x}_t, X_0 = \mathbf{x}_0) &= \frac{q(X_t = \mathbf{x}_t|X_{t-1}, X_0 = \mathbf{x}_0)q(X_{t-1}|X_0 = \mathbf{x}_0)}{q(X_t = \mathbf{x}_t|X_0 = \mathbf{x}_0)} \\
 &= \frac{q(X_t = \mathbf{x}_t|X_{t-1})q(X_{t-1}|X_0 = \mathbf{x}_0)}{q(X_t = \mathbf{x}_t|X_0 = \mathbf{x}_0)}
 \end{aligned}$$

$$\underbrace{q(X_{t-1}|X_0 = \mathbf{x}_0)}_{1 \times k} = \underbrace{\mathbf{x}_0}_{1 \times k} \underbrace{\bar{Q}_{t-1}}_{k \times k}$$

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$$\begin{aligned}
 q(X_{t-1} | X_t = \mathbf{x}_t, X_0 = \mathbf{x}_0) &= \frac{q(X_t = \mathbf{x}_t | X_{t-1}, X_0 = \mathbf{x}_0) q(X_{t-1} | X_0 = \mathbf{x}_0)}{q(X_t = \mathbf{x}_t | X_0 = \mathbf{x}_0)} \\
 &= \frac{q(X_t = \mathbf{x}_t | X_{t-1}) q(X_{t-1} | X_0 = \mathbf{x}_0)}{q(X_t = \mathbf{x}_t | X_0 = \mathbf{x}_0)}
 \end{aligned}$$

$$\underbrace{q(X_t = \mathbf{x}_t | X_0 = \mathbf{x}_0)}_{1 \times 1} = \underbrace{\begin{bmatrix} \mathbf{x}_0 & \bar{Q}_t \end{bmatrix}}_{\substack{1 \times k & k \times k}} \underbrace{\mathbf{x}_t^T}_{k \times 1}$$

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$$q(X_{t-1}|X_t = \mathbf{x}_t, X_0 = \mathbf{x}_0) = \text{Cat}\left(\mathbf{x}_{t-1}; \frac{\underbrace{\mathbf{x}_t Q_t^T}_{\text{vector}} \odot \underbrace{\mathbf{x}_0 \bar{Q}_{t-1}}_{\text{vector}}}{\underbrace{\mathbf{x}_0 \bar{Q}_t \mathbf{x}_t^T}_{\text{scalar}}}\right).$$

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- Reverse process:
 - Two key distributions: $q(\mathbf{x}_0|\mathbf{x}_t)$ and $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$
 - Approximate $q(\mathbf{x}_0|\mathbf{x}_t)$ with $p_\theta(\mathbf{x}_0|\mathbf{x}_t)$.

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{\sum_{\mathbf{x}_0} q(\mathbf{x}_{t-1}, \mathbf{x}_t, \mathbf{x}_0)}{q(\mathbf{x}_t)} \quad (7)$$

$$= \frac{\sum_{\mathbf{x}_0} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)q(\mathbf{x}_0|\mathbf{x}_t)q(\mathbf{x}_t)}{q(\mathbf{x}_t)} \quad (7b)$$

$$= \sum_{\mathbf{x}_0} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)q(\mathbf{x}_0|\mathbf{x}_t) \quad (7c)$$

$$= \mathbb{E}_{q(\mathbf{x}_0|\mathbf{x}_t)} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \quad (7d)$$

$$= \mathbb{E}_{p_\theta(\mathbf{x}_0|\mathbf{x}_t)} q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \quad (7d)$$

Diffusion Model for Discrete Data

- Discrete Diffusion Model for language modeling

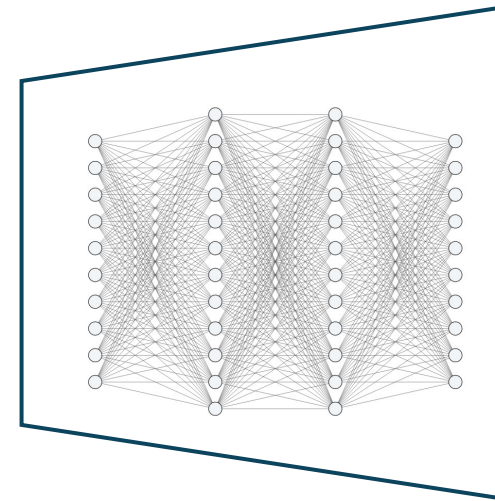
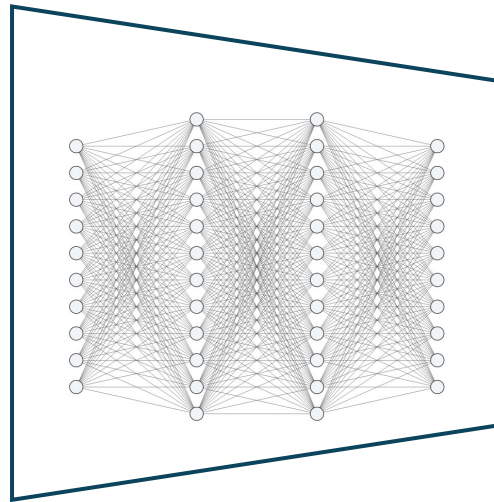
$$Q_i^{\text{uniform}} = \begin{matrix} & \begin{matrix} M \times M \end{matrix} \\ \begin{bmatrix} 1 - \beta_i + \beta_i/M & \beta_i/M & \cdots & \beta_i/M \\ \beta_i/M & 1 - \beta_i + \beta_i/M & \cdots & \beta_i/M \\ \vdots & \vdots & \ddots & \vdots \\ \beta_i/M & \beta_i/M & \cdots & 1 - \beta_i + \beta_i/M \end{bmatrix} \end{matrix}$$

$$Q_i^{\text{mask}} = \begin{matrix} & \begin{matrix} (M+1) \times (M+1) \end{matrix} \\ \begin{bmatrix} 1 - \beta_i & 0 & \cdots & 0 & \beta_i \\ 0 & 1 - \beta_i & \cdots & 0 & \beta_i \\ \vdots & \vdots & \ddots & 0 & \beta_i \\ 0 & 0 & \cdots & 1 - \beta_i & \beta_i \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix} \end{matrix}$$



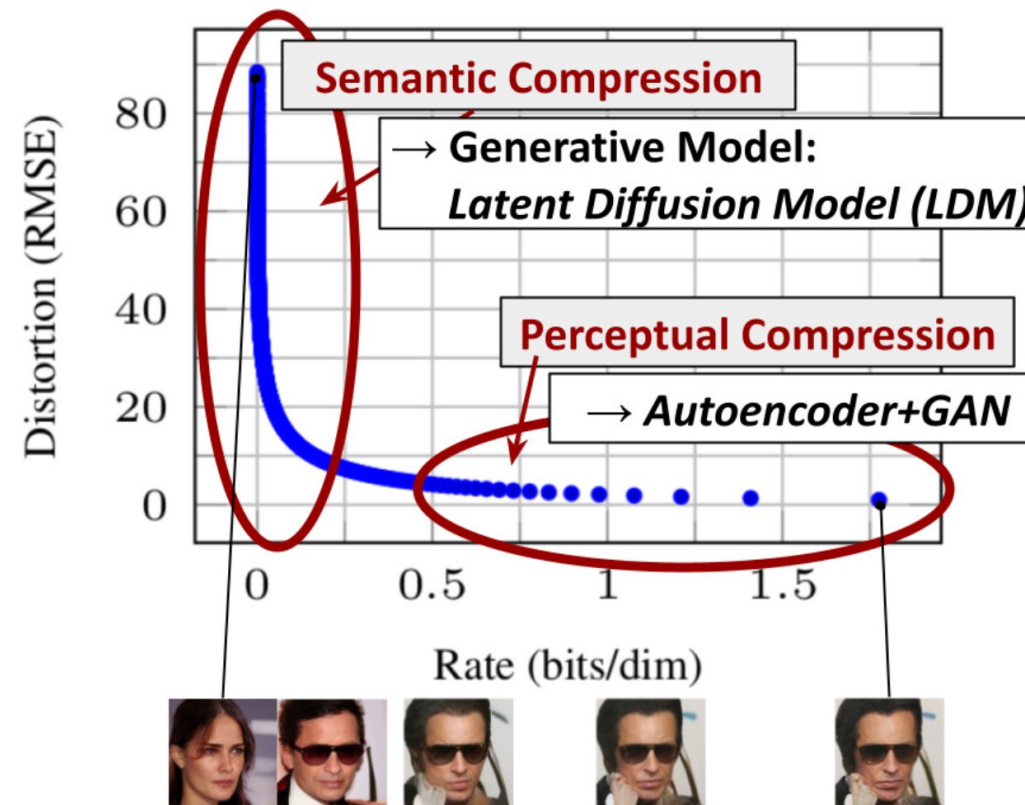
Latent Diffusion Model

- Combine with VAE, run the diffusion process in **the latent space** instead of pixel space



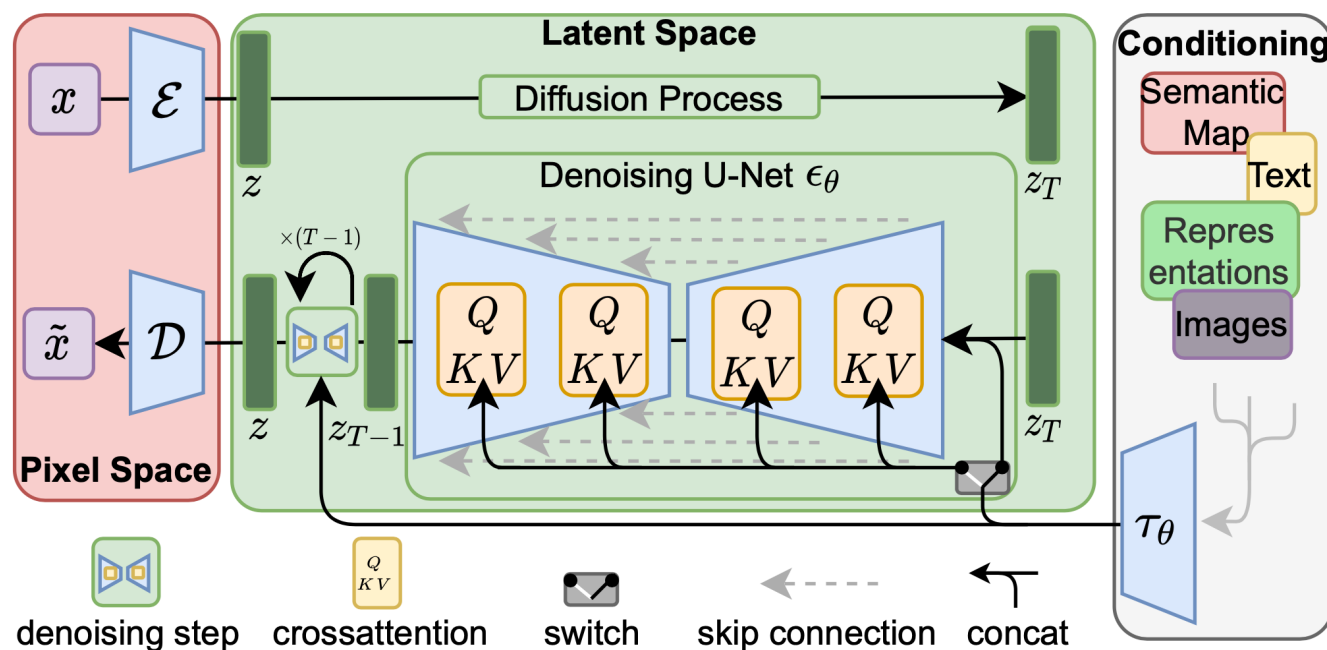
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Latent Diffusion Model

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CompVis/stable-diffusion

A latent text-to-image diffusion model



8 Contributors 536 Issues 73k Stars 11k Forks





To be continued